KITABU QUEST

Form I

COMPUTER SCIENCE

Giraffe

Cover scene

students using computers safely with a giraffe mascot

- Clear lessons
- Worked examples
- Fun activities
- Real-life problems

GRADE

I

TZA TIE-BASIC aligned draft

Think - Explore - Achieve

Title Page

KITABU QUEST Form I Computer Science

Tanzania TIE-BASIC learner book

Mascot: giraffe

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Tanzania learners.

How To Use This Book

This book helps Form I learners study Computer Science one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Practise safe computing, algorithms, data, systems thinking, and clear step-by-step explanations.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

As you finish each topic, return to the success criteria and correct one answer before moving on.

Computer Systems Hardware And Software: Lesson Opener

Learning focus: Computer Systems Hardware And Software

Country link: a safe Tanzania observation, model, data table, or investigation for Computer Systems Hardware And Software.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- investigate and explain computer systems hardware and software using evidence

Inquiry questions:
- How can computer systems hardware and software help you solve a real problem?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Computer Systems Hardware And Software: Learn

Computer Systems Hardware And Software is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:
- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:
- Computer Systems Hardware And Software
- Computer
- Systems
- Hardware
- Software
- concept
- observation
- evidence

Computer Systems Hardware And Software: Worked Example

Worked example for Computer Systems Hardware And Software: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Computer Systems Hardware And Software: Vocabulary And Study Skill

Use these words and study moves before you practise Computer Systems Hardware And Software. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Computer Systems Hardware And Software
- Computer
- Systems
- Hardware
- Software
- concept
- observation
- evidence

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Computer Systems Hardware And Software without a clear example, method, or evidence.

Computer Systems Hardware And Software: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Computer Systems Hardware And Software.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Computer Systems Hardware And Software: Practice And Correction

Practice for Computer Systems Hardware And Software:

Main outcome: investigate and explain computer systems hardware and software using evidence.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Computer Systems Hardware And Software: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Computer Systems Hardware And Software.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Computer Systems Hardware And Software: Outcome Check

Outcome check for Computer Systems Hardware And Software: Investigate and explain computer systems hardware and software using evidence.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Input Output Storage And Device Care: Lesson Opener

Learning focus: Input Output Storage And Device Care

Country link: a safe Tanzania observation, model, data table, or investigation for Input Output Storage And Device Care.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- investigate and explain input output storage and device care using evidence

Inquiry questions:
- How can input output storage and device care help you solve a real problem?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Input Output Storage And Device Care: Learn

Input Output Storage And Device Care is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:
- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:
- Input Output Storage And Device Care
- Input
- Output
- Storage
- Device
- Care
- concept
- observation

Input Output Storage And Device Care: Worked Example

Investigation model for Input Output Storage And Device Care: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Input Output Storage And Device Care: Vocabulary And Study Skill

Use these words and study moves before you practise Input Output Storage And Device Care. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Input Output Storage And Device Care
- Input
- Output
- Storage
- Device
- Care
- concept
- observation

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Input Output Storage And Device Care without a clear example, method, or evidence.

Input Output Storage And Device Care: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Input Output Storage And Device Care.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Input Output Storage And Device Care: Practice And Correction

Practice for Input Output Storage And Device Care:

Main outcome: investigate and explain input output storage and device care using evidence.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Input Output Storage And Device Care: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Input Output Storage And Device Care.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Input Output Storage And Device Care: Outcome Check

Outcome check for Input Output Storage And Device Care: Investigate and explain input output storage and device care using evidence.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Operating Systems And File Management: Lesson Opener

Learning focus: Operating Systems And File Management

Country link: a safe Tanzania observation, model, data table, or investigation for Operating Systems And File Management.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- investigate and explain operating systems and file management using evidence

Inquiry questions:
- How can operating systems and file management help you solve a real problem?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Operating Systems And File Management: Learn

Operating Systems And File Management is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:
- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:
- Operating Systems And File Management
- Operating
- Systems
- File
- Management
- concept
- observation
- evidence

Operating Systems And File Management: Worked Example

Worked example for Operating Systems And File Management: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Operating Systems And File Management: Vocabulary And Study Skill

Use these words and study moves before you practise Operating Systems And File Management. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Operating Systems And File Management
- Operating
- Systems
- File
- Management

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Operating Systems And File Management without a clear example, method, or evidence.

Operating Systems And File Management: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Operating Systems And File Management.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Operating Systems And File Management: Practice And Correction

Practice for Operating Systems And File Management:

Main outcome: investigate and explain operating systems and file management using evidence.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Operating Systems And File Management: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Operating Systems And File Management.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Operating Systems And File Management: Outcome Check

Outcome check for Operating Systems And File Management: Investigate and explain operating systems and file management using evidence.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Word Processing For School Work: Lesson Opener

Learning focus: Word Processing For School Work

Country link: a safe Tanzania observation, model, data table, or investigation for Word Processing For School Work.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- investigate and explain word processing for school work using evidence

Inquiry questions:
- How can word processing for school work help you solve a real problem?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Word Processing For School Work: Learn

Word Processing For School Work teaches useful computer work through exact steps, not guessing.

Core idea:
- Word processing is used to create, edit, format, save, and print text documents.
- A spreadsheet organizes data in rows, columns, and cells.
- A cell can hold text, a number, or a formula.
- Formatting should make work clearer, not just more decorative.
- Good digital work is saved with a clear file name and checked before sharing.

Learner task link: create a class list, simple budget, marks table, timetable, or short report using safe school devices.

Key words:
- Word Processing For School Work
- Word
- Processing
- School
- Work
- concept
- observation
- evidence

Word Processing For School Work: Worked Example

Worked example for Word Processing For School Work: Make a simple spreadsheet for class garden harvest.

1. Open a blank worksheet.
2. In row 1, type headings: Crop, Week 1, Week 2, Total.
3. Enter beans, maize, and tomatoes in the Crop column.
4. Type harvest numbers for Week 1 and Week 2.
5. In the Total column, use a formula such as =B2+C2.
6. Save the file with a clear name, for example garden-harvest-p6.

Quality check: headings are clear, numbers are in the correct cells, formulas give sensible totals, and the file is saved.

Word Processing For School Work: Vocabulary And Study Skill

Use these words and study moves before you practise Word Processing For School Work. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Word Processing For School Work
- Word
- Processing
- School
- Work
- concept
- observation
- evidence

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Word Processing For School Work without a clear example, method, or evidence.

Word Processing For School Work: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Word Processing For School Work.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Word Processing For School Work: Practice And Correction

Practice for Word Processing For School Work:

Main outcome: investigate and explain word processing for school work using evidence.

1. Create a document or worksheet with a clear title and headings.
2. Enter five rows of useful class, garden, timetable, budget, or marks data.
3. Format the work neatly, save it with a clear file name, and explain one formula or formatting choice.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Word Processing For School Work: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Word Processing For School Work.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Word Processing For School Work: Outcome Check

Outcome check for Word Processing For School Work: Investigate and explain word processing for school work using evidence.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Internet Safety And Responsible Use: Lesson Opener

Learning focus: Internet Safety And Responsible Use

Country link: a safe Tanzania observation, model, data table, or investigation for Internet Safety And Responsible Use.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- investigate and explain internet safety and responsible use using evidence

Inquiry questions:
- How can internet safety and responsible use help you solve a real problem?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Internet Safety And Responsible Use: Learn

Internet Safety And Responsible Use begins with safe behaviour and accurate measurement.

Core idea:
- Physics uses measurements such as length, mass, time, temperature, and volume.
- A measurement is useful only when the number and unit are both written.
- Safety rules protect learners, equipment, and results.
- Read scales at eye level and record values immediately.

Local link: in Tanzania, careful measurement helps in health, building, farming, weather records, transport, and school laboratory work.

Key words:
- Internet Safety And Responsible Use
- Internet
- Safety
- Responsible
- concept
- observation
- evidence
- safety

Internet Safety And Responsible Use: Worked Example

Investigation model for Internet Safety And Responsible Use: Measure the length of a desk safely.

1. Choose a metre rule or tape measure.
2. Place zero at one end of the desk.
3. Read the scale at eye level.
4. Record the value with the unit, for example 120 cm.
5. Repeat once and compare readings.

Conclusion: a useful measurement includes a number, a unit, and a careful method.

Internet Safety And Responsible Use: Vocabulary And Study Skill

Use these words and study moves before you practise Internet Safety And Responsible Use. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Internet Safety And Responsible Use
- Internet
- Safety
- Responsible
- concept
- observation
- evidence
- safety

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Internet Safety And Responsible Use without a clear example, method, or evidence.

Internet Safety And Responsible Use: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Internet Safety And Responsible Use.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Internet Safety And Responsible Use: Practice And Correction

Practice for Internet Safety And Responsible Use:

Main outcome: investigate and explain internet safety and responsible use using evidence.

1. Name three laboratory safety rules.
2. Measure one classroom object and record the value with a unit.
3. Explain one error that can make a measurement unreliable.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Internet Safety And Responsible Use: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Internet Safety And Responsible Use.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Internet Safety And Responsible Use: Outcome Check

Outcome check for Internet Safety And Responsible Use: Investigate and explain internet safety and responsible use using evidence.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Algorithms As Step-By-Step Instructions: Lesson Opener

Learning focus: Algorithms As Step-By-Step Instructions

Country link: a safe Tanzania observation, model, data table, or investigation for Algorithms As Step-By-Step Instructions.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- investigate and explain algorithms as step-by-step instructions using evidence

Inquiry questions:
- How can algorithms as step-by-step instructions help you solve a real problem?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Algorithms As Step-By-Step Instructions: Learn

Algorithms As Step-By-Step Instructions is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:
- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:
- Algorithms As Step-By-Step Instructions
- Algorithms
- Step-By-Step
- Instructions
- concept
- observation
- evidence
- safety

Algorithms As Step-By-Step Instructions: Worked Example

Worked example for Algorithms As Step-By-Step Instructions: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Algorithms As Step-By-Step Instructions: Vocabulary And Study Skill

Use these words and study moves before you practise Algorithms As Step-By-Step Instructions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Algorithms As Step-By-Step Instructions
- Algorithms
- Step-By-Step
- Instructions
- concept
- observation
- evidence
- safety

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Algorithms As Step-By-Step Instructions without a clear example, method, or evidence.

Algorithms As Step-By-Step Instructions: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Algorithms As Step-By-Step Instructions.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Algorithms As Step-By-Step Instructions: Practice And Correction

Practice for Algorithms As Step-By-Step Instructions:

Main outcome: investigate and explain algorithms as step-by-step instructions using evidence.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Algorithms As Step-By-Step Instructions: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Algorithms As Step-By-Step Instructions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Algorithms As Step-By-Step Instructions: Outcome Check

Outcome check for Algorithms As Step-By-Step Instructions: Investigate and explain algorithms as step-by-step instructions using evidence.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Data Tables And Simple Charts: Lesson Opener

Learning focus: Data Tables And Simple Charts

Country link: a safe Tanzania observation, model, data table, or investigation for Data Tables And Simple Charts.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- investigate and explain data tables and simple charts using evidence

Inquiry questions:
- How can data tables and simple charts help you solve a real problem?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Data Tables And Simple Charts: Learn

Data Tables And Simple Charts is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:
- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:
- Data Tables And Simple Charts
- Data
- Tables
- Simple
- Charts
- concept
- observation
- evidence

Data Tables And Simple Charts: Worked Example

Investigation model for Data Tables And Simple Charts: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Data Tables And Simple Charts: Vocabulary And Study Skill

Use these words and study moves before you practise Data Tables And Simple Charts. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Data Tables And Simple Charts
- Data
- Tables
- Simple
- Charts
- concept
- observation
- evidence

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Data Tables And Simple Charts without a clear example, method, or evidence.

Data Tables And Simple Charts: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Data Tables And Simple Charts.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Data Tables And Simple Charts: Practice And Correction

Practice for Data Tables And Simple Charts:

Main outcome: investigate and explain data tables and simple charts using evidence.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Data Tables And Simple Charts: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Data Tables And Simple Charts.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Data Tables And Simple Charts: Outcome Check

Outcome check for Data Tables And Simple Charts: Investigate and explain data tables and simple charts using evidence.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Form I Computing Project: Lesson Opener

Learning focus: Form I Computing Project

Country link: a safe Tanzania observation, model, data table, or investigation for Form I Computing Project.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- investigate and explain form i computing project using evidence

Inquiry questions:
- How can form i computing project help you solve a real problem?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Form I Computing Project: Learn

Form I Computing Project is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:
- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:
- Form I Computing Project
- Form
- Computing
- Project
- concept

Form I Computing Project: Worked Example

Investigation model for Form I Computing Project: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Form I Computing Project: Vocabulary And Study Skill

Use these words and study moves before you practise Form I Computing Project. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Form I Computing Project
- Form
- Computing
- Project
- concept
- observation
- evidence
- safety

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Form I Computing Project without a clear example, method, or evidence.

Form I Computing Project: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Form I Computing Project.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Form I Computing Project: Practice And Correction

Practice for Form I Computing Project:

Main outcome: investigate and explain form i computing project using evidence.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Form I Computing Project: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Form I Computing Project.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Form I Computing Project: Outcome Check

Outcome check for Form I Computing Project: Investigate and explain form i computing project using evidence.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Computer Systems Hardware And Software: Review Clinic 1

Use this review clinic to strengthen Computer Science without copying earlier answers.

1. Choose one idea from Computer Systems Hardware And Software that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Input Output Storage And Device Care: Review Clinic 2

Use this review clinic to strengthen Computer Science without copying earlier answers.

1. Choose one idea from Input Output Storage And Device Care that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Operating Systems And File Management: Review Clinic 3

Use this review clinic to strengthen Computer Science without copying earlier answers.

1. Choose one idea from Operating Systems And File Management that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Word Processing For School Work: Review Clinic 4

Use this review clinic to strengthen Computer Science without copying earlier answers.

1. Choose one idea from Word Processing For School Work that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Computer Systems Hardware And Software: Application Workshop 1

Application workshop 1 for Computer Systems Hardware And Software.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Input Output Storage And Device Care: Application Workshop 2

Application workshop 2 for Input Output Storage And Device Care.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Operating Systems And File Management: Application Workshop 3

Application workshop 3 for Operating Systems And File Management.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Word Processing For School Work: Application Workshop 4

Application workshop 4 for Word Processing For School Work.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Internet Safety And Responsible Use: Application Workshop 5

Application workshop 5 for Internet Safety And Responsible Use.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Algorithms As Step-By-Step Instructions: Application Workshop 6

Application workshop 6 for Algorithms As Step-By-Step Instructions.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Data Tables And Simple Charts: Application Workshop 7

Application workshop 7 for Data Tables And Simple Charts.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Form I Computing Project: Application Workshop 8

Application workshop 8 for Form I Computing Project.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Computer Systems Hardware And Software: Application Workshop 9

Application workshop 9 for Computer Systems Hardware And Software.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Input Output Storage And Device Care: Application Workshop 10

Application workshop 10 for Input Output Storage And Device Care.

Grade move: build lower-secondary foundations with clear vocabulary and guided application; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Glossary

Use this glossary to revise important words in Computer Science.

- Computer Systems Hardware And Software: Explain this term using an example from the book, your school, or your community.
- Input Output Storage And Device Care: Explain this term using an example from the book, your school, or your community.
- Operating Systems And File Management: Explain this term using an example from the book, your school, or your community.
- Word Processing For School Work: Explain this term using an example from the book, your school, or your community.
- Internet Safety And Responsible Use: Explain this term using an example from the book, your school, or your community.
- Algorithms As Step-By-Step Instructions: Explain this term using an example from the book, your school, or your community.
- Data Tables And Simple Charts: Explain this term using an example from the book, your school, or your community.
- Form I Computing Project: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Computer Science answer states the idea, uses correct scientific vocabulary, records observations or data, includes units or labels where needed, and writes a conclusion that follows from evidence. Safety rules always come first. If evidence is missing, repeat the observation safely or explain what evidence would be needed.

Final Project

Create a final project for Computer Science that shows what you can now do independently.

Your project must include:
1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.